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**Department of Artificial Intelligence**  
**Probabilistic Graphical Models (AI453)**  
**[Lab Practical - 5]**

**Problem Statement:** Design, Estimation and Inference of a **Bayesian Network** learned from **Real Secondary-School Performance Data**, including exact inference by enumeration, ancestral and rejection sampling, and model criticism.

## INTRODUCTION

Everything so far has been given to you. Practical 1 and Practical 2 supplied a joint distribution as eight numbers and asked you to manipulate it. Practical 3 supplied distributions and asked you to sample from them. In each case the distribution existed before the exercise began.

A Bayesian network is not a table of numbers. It is a directed acyclic graph together with one conditional probability table per node, and the claim it makes is that the full joint distribution is the product of those tables. Eight binary variables would otherwise require 255 free parameters; the network in this practical requires 40. That reduction is the entire practical value of the graph.

**Key Insight:** Estimating a conditional probability table from data is counting, and nothing more.  $P(v = 1 \mid \text{parents})$  is the number of rows in which  $v$  is 1 and the parents take the given values, divided by the number of rows in which the parents take those values. This is the same maximum likelihood estimate you computed for a loaded die in Practical 3, applied once per parent configuration. Its weakness is also the same: when a parent configuration is rare, the estimate is unstable, and when it never occurs at all the estimate does not exist.

The graph in this practical was not learned from the data. It was written down from the order in which things happen to a student: family background first, then behaviour settled early in the year, then the first-period grade, then the final grade. A graph asserts conditional independences, the data either supports them or does not, and comparing the two is a required part of the work rather than an optional extra.

## DATASET

`student-por.csv` — records of 649 students of a Portuguese language course at two secondary schools, collected by Cortez and Silva (2008) and held in the UCI Machine Learning Repository. The file is supplied with this practical; no download and no network access are required.

Eight columns are reduced to binary variables. The joint distribution therefore has  $2^8 = 256$  entries and remains small enough to enumerate exhaustively, which is what makes exact verification possible throughout.

| Variable | Meaning                               | Cut from the column | $P(v = 1)$ |
|----------|---------------------------------------|---------------------|------------|
| medu     | mother educated to secondary or above | Medu $\geq 3$       | 0.484      |
| goout    | goes out with friends often           | goout $\geq 4$      | 0.387      |
| fail     | has failed a class before             | failures $\geq 1$   | 0.154      |
| higher   | wants to pursue higher education      | higher == yes       | 0.894      |

|        |                                  |                    |       |
|--------|----------------------------------|--------------------|-------|
| study  | studies more than 5 hours a week | studytime $\geq 3$ | 0.203 |
| absent | six or more absences in the year | absences $\geq 6$  | 0.263 |
| g1     | first-period grade of 12 or more | G1 $\geq 12$       | 0.471 |
| grade  | final grade of 12 or more        | G3 $\geq 12$       | 0.536 |

## THE NETWORK

The structure is supplied in the starter file as a dictionary of child to parents. Confirm before relying on it that the variable list is a topological order, that is, every variable appears after all of its parents.

| Node   | Parents                    | Conditional probability table |
|--------|----------------------------|-------------------------------|
| medu   | —                          | 1 row                         |
| goout  | —                          | 1 row                         |
| fail   | medu                       | 2 rows                        |
| higher | medu, fail                 | 4 rows                        |
| study  | medu, higher, goout        | 8 rows                        |
| absent | goout, fail                | 4 rows                        |
| g1     | fail, study, goout, higher | 16 rows                       |
| grade  | g1, absent                 | 4 rows                        |

**Constraints:** Python 3 standard library only. No NumPy, no pandas, no pgmpy. Run `lab5_bn.py`; it prints the worked example  $P(\text{fail}=1 \mid \text{medu}=1) = 0.099$ . The loader and the counting helper are supplied; everything else is to be written.

### Problem Statement 1: Data Preparation and Marginals

Load the file, apply the binary cuts and establish that the data is what the table above claims.

#### Tasks:

- Read the CSV with the supplied `load()` helper and report the number of records; it must be 649
- Compute and print  $P(v = 1)$  for all eight variables and check them against the dataset table
- For each of the three continuous columns that were thresholded, state what is lost by binarising, and identify the one variable whose marginal is extreme enough to cause difficulty later

### Problem Statement 2: Parameter Estimation by Counting

Fit every conditional probability table from the data by maximum likelihood.

#### Estimator:

$$P(v = 1 \mid \text{pa}) = (n1 + \text{alpha}) / (n0 + n1 + 2 * \text{alpha})$$

n1 rows with  $v = 1$  and the parents at the given configuration  
 n0 rows with  $v = 0$  and the parents at the given configuration  
 alpha smoothing constant; 0 for plain maximum likelihood

### Tasks:

- d) Implement `fit(data, alpha=0.0)` returning `cpt[v][parent_tuple] = P(v = 1 | parents)`
- e) Print the table of `grade`: four rows, one per (g1, absent) pair, each with the number of students supporting it
- f) State which of the two parents of `grade` carries more information, and quantify the claim from your own four rows
- g) Count the total number of free parameters in the fitted network and compare with the 255 that an unrestricted distribution over eight binary variables would require

### Problem Statement 3: Factorization of the Joint Distribution

The product over nodes of  $P(v \mid \text{parents})$  is the whole content of the graph. Verify it is a distribution.

### Tasks:

- h) Implement `joint(assignment, cpt)` returning the probability of one complete assignment to all eight variables as the product of the eight local terms
- i) Enumerate all 256 assignments and confirm the total is 1 to within floating-point tolerance
- j) If the total is not 1, the fault is a conditional probability row that does not sum to 1 or a parent lookup performed in the wrong order. Describe how you would distinguish the two cases from the output alone

### Problem Statement 4: Exact Inference by Enumeration

Answer conditional queries by summing the joint distribution, and validate the answers against the raw data. This routine is slow and exponential, and it is the reference against which every faster algorithm later in the course is checked.

### Tasks:

- k) Implement `infer(query, evidence, cpt): sum joint()` over every assignment consistent with the evidence, accumulate a second total over those that also match the query, and divide
- l) Compute the following, printing beside each the plain fraction observed in the data and the number of students that fraction rests on:
 

|                                                  |                                                |
|--------------------------------------------------|------------------------------------------------|
| <code>P(grade = 1)</code>                        | <code>P(grade = 1   g1 = 1)</code>             |
| <code>P(grade = 1   fail = 0)</code>             | <code>P(grade = 1   g1 = 1, absent = 1)</code> |
| <code>P(grade = 1   study = 1, goout = 0)</code> | <code>P(grade = 1   fail = 1)</code>           |
- m) Five of the six agree closely. Identify the one that does not and set it aside for Problem Statement 8
- n) State the number of assignments the routine visits as a function of the number of variables, and the point at which this approach becomes impractical

Read the independences off the graph, then test them against the data.

- a) Three pairs of variables have neither an edge between them nor a common ancestor, so the graph asserts they are independent outright: `medu` and `goout`, `goout` and `fail`, `goout` and `higher`. For each, print  $P(a = 1, b = 1)$  beside  $P(a = 1) P(b = 1)$  and state whether the data supports the assertion
- b) `medu` and `goout` are both parents of `g1`. Conditioning on a collider makes independent variables dependent. Print

$P(\text{medu} = 1)$   
 $P(\text{medu} = 1 \mid \text{g1} = 0, \text{goout} = 0)$

$P(\text{medu} = 1 \mid \text{g1} = 0)$   
 $P(\text{medu} = 1 \mid \text{g1} = 0, \text{goout} = 1)$
- c) Explain, in terms of competing causes, why learning that a low-scoring student goes out frequently makes an educated mother *more* likely rather than less. Relate the mechanism to the sprinkler and rain example of Practical 2
- d) Repeat the collider check conditioning on `grade` rather than on `g1`, and explain why conditioning on a descendant of a collider has the same effect
- e) Identify one variable for which the graph asserts an independence that the data does not support, and propose the edge that would repair it

The table of g1 has four parents, so sixteen rows must be filled from 649 students. Some rows are supported by very few records and at least one by none at all.

- t) Fit with  $\alpha = 0$  and list every row that came out as exactly 0, exactly 1, or as an undefined 0/0, giving the supporting count for each
- u) Explain what a row of exactly 0 does to any query in Problem Statement 4 that requires it, and what an undefined row does
- v) Refit with  $\alpha = 1$  and confirm every degenerate row is gone. Relate the effect to the die fitted on thirty rolls in Practical 3, where face 6 was assigned probability zero
- w) Report  $P(\text{grade} = 1 \mid g1 = 1)$  for  $\alpha$  in  $\{0.01, 0.1, 0.5, 1.0, 2.0\}$  and describe the trade-off that governs the choice of  $\alpha$

Generate students from the fitted network and use them to answer the same queries approximately.

- x) Implement `forward(cpt)` producing one simulated student. Walk the variables in topological order; each is a `sample_bernoulli()` draw with `p` read from its own table using the values already drawn for its parents
- y) State precisely why the topological order is required, and what fails if a variable is drawn before one of its parents

- z) Implement rejection sampling: draw  $M$  students, discard those inconsistent with the evidence, and take the fraction of the survivors that match the query. Estimate  $P(\text{grade} = 1 \mid \text{fail} = 1)$  at  $M = 2000$  and  $M = 50000$  against the exact value from Problem Statement 4, and report the percentage of samples retained
- aa) Repeat with the evidence  $g1 = 0$ ,  $goout = 1$ ,  $study = 1$ ,  $higher = 0$ . Report the retention rate and the resulting estimate at both values of  $M$
- ab) Explain why the retention rate collapses as evidence is added, why this makes rejection sampling unusable on any network of realistic size, and name the family of algorithms introduced to replace it

### Problem Statement 8: Model Criticism

The graph encodes a story about how school works. Determine where the 649 students disagree with that story.

#### Tasks:

- ac) Print the model estimate of  $P(\text{grade} = 1 \mid \text{fail} = 1)$  beside the observed fraction. Report the discrepancy
- ad) Refit at  $\alpha = 0.01, 0.1, 1.0$  and  $2.0$  and demonstrate that the discrepancy is not attributable to smoothing
- ae) Trace every directed path from fail to grade in the graph and explain why each transmits only weakly. Name the edge whose absence accounts for the discrepancy, add it, refit and report the corrected value
- af) The variable absent is recorded over the whole academic year while  $g1$  is fixed at the midpoint. Explain why this makes the assertion that  $g1$  separates absent from grade doubtful, and state what a correct treatment would require
- ag) Conclude in three or four lines: the structure was assumed rather than learned, and the data contradicted it in specific places. State what the appropriate remedy is and name the topic in this course that provides it

### DELIVERABLES

A single Python file containing all eight problem statements, together with its printed output and the written answers placed as comments beside the task requiring them.

### DATASET LINKS

**1. UCI Student Performance:** <https://archive.ics.uci.edu/dataset/320/student+performance>  
*Supplied with this practical as student-por.csv; no download is required.*

### REFERENCES

1. D. Koller and N. Friedman, Probabilistic Graphical Models: Principles and Techniques, MIT Press, Chapters 3 and 17.
2. A. Darwiche, Modeling and Reasoning with Bayesian Networks, Chapters 5 and 17.
3. K. Murphy, Machine Learning: A Probabilistic Perspective, Chapter 10.

4. P. Cortez and A. Silva, Using Data Mining to Predict Secondary School Student Performance, Proceedings of 5th FUBUTEC, 2008. UCI Machine Learning Repository.